

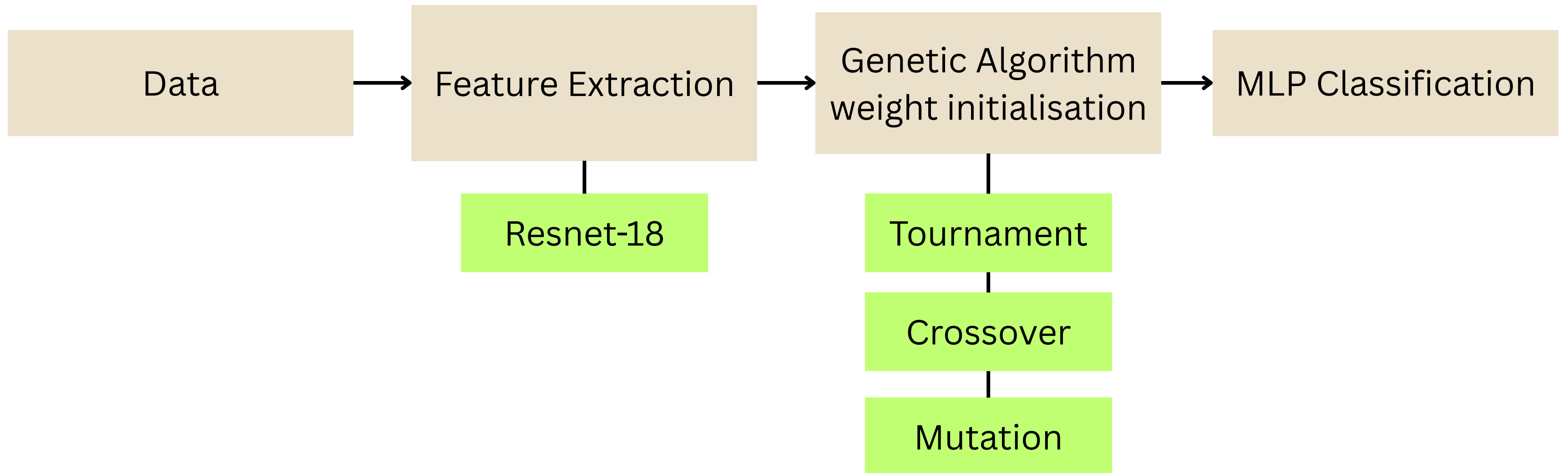
Initialize the weights of a multilayer perceptron network using the genetic algorithm and classify a satellite image with the weight initialized neural network

by

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Plan of Action



Dataset : EuroSAT

Public labelled dataset of Sentinel-2 Images

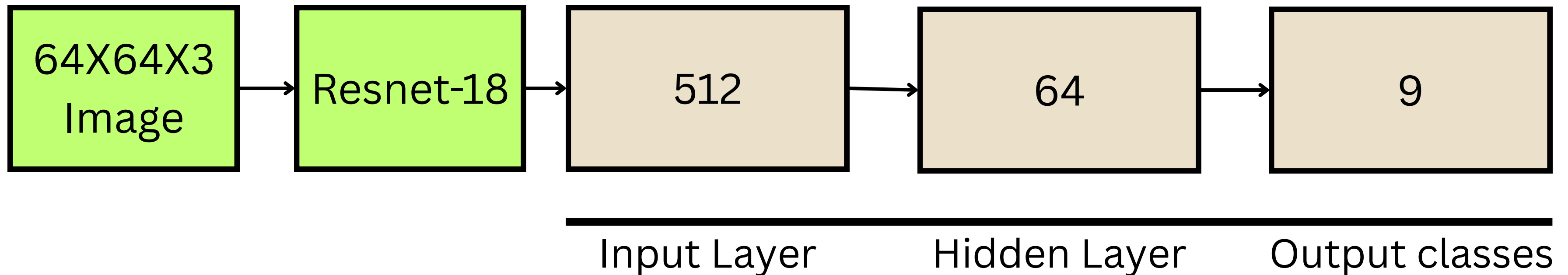
Count of data: 24500

No of classes: 10 - 1 (*River class dropped. Too many missclassifications*)

Classes: 'AnnualCrop', 'Forest', 'HerbaceousVegetation', 'Highway', 'Industrial',
'Pasture', 'PermanentCrop', 'Residential', 'SeaLake'

MLP

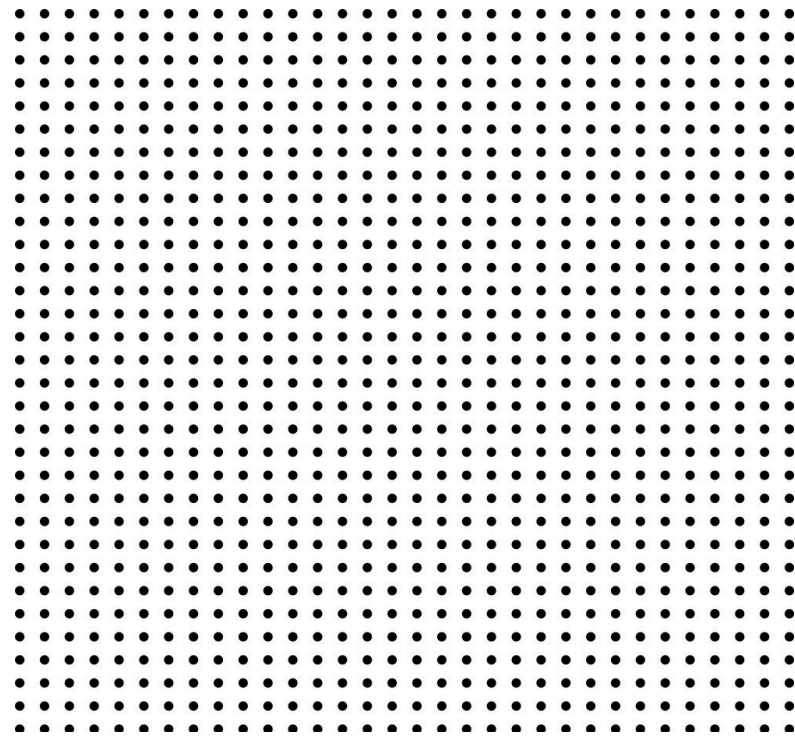
Input Feature Extractor



MLP

Genetic Algorithm weight initialisation

Step -1 Population Initialisation



Hyperparameter-1: Population size 100

Individual in Population



Input size * HL1 + HL1 size + HL1 size * output size

WM -1

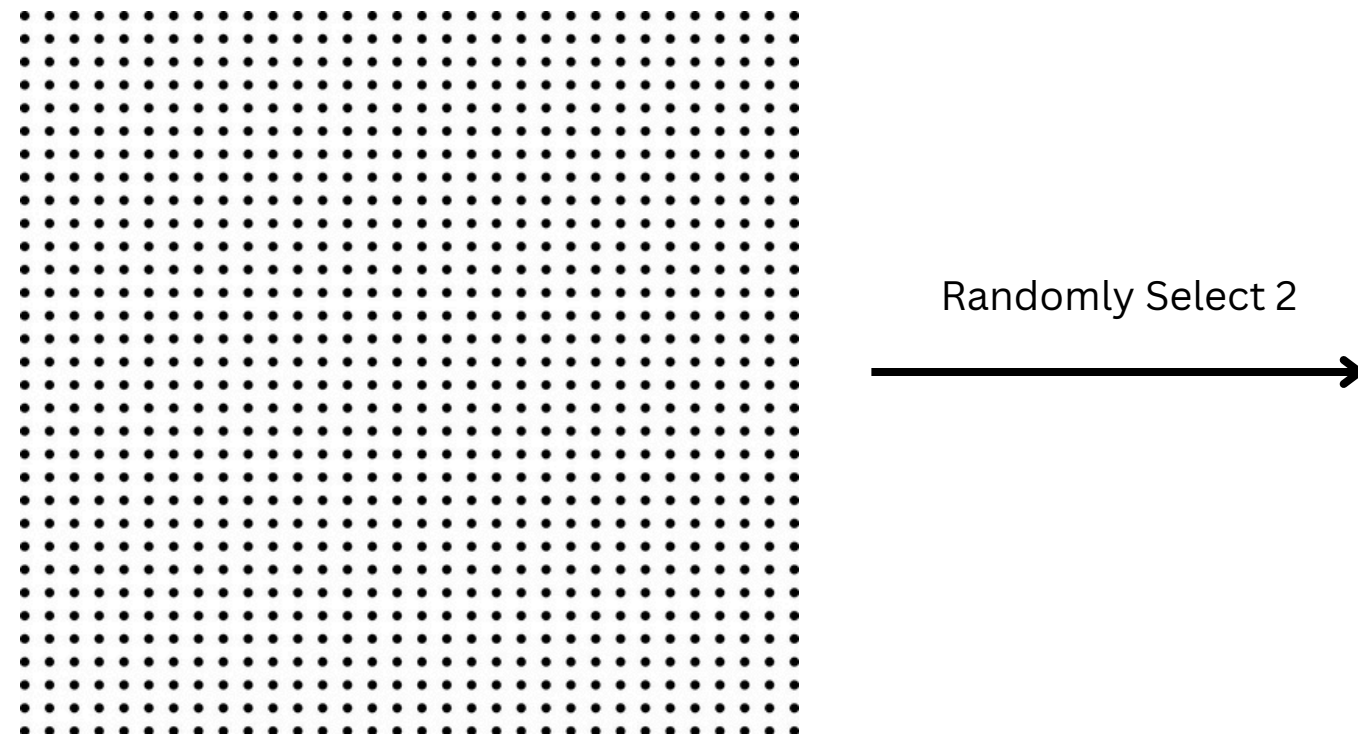
Bias

WM -2

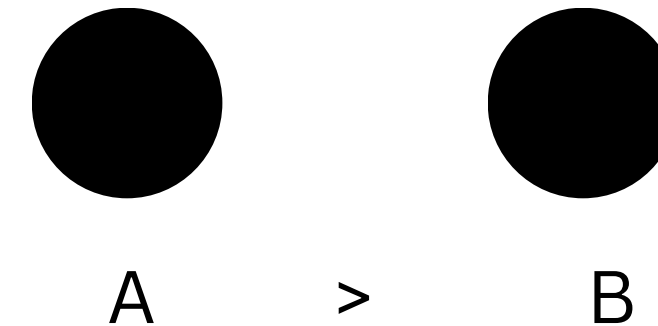
WM- Weight Matrix
HL- Hidden layer size

Genetic Algorithm weight initialisation

Step -2 Tournament Selection



Compare Fitness



Winner goes to next gen population set
Done twice for each population choosing iteration

Hyperparameter-2: Tournament size 2

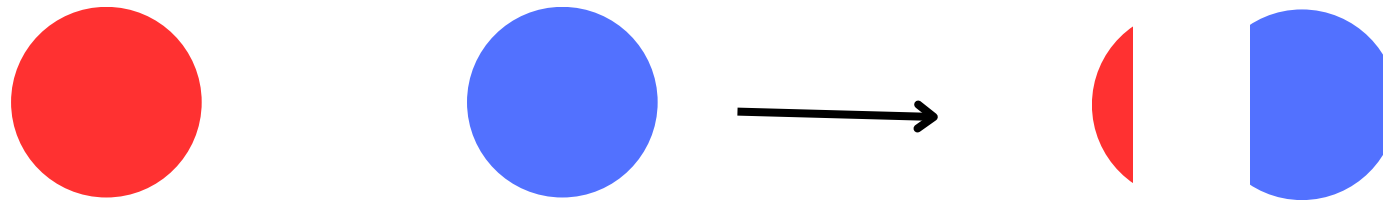
Fitness Function: Accuracy from MLP classification.

Genetic Algorithm weight initialisation

Step -3 Crossover

Step -4 Mutation

Parent genes selected from previous step



Crossed from a prandom point in Genes

New Parent

Randomly chosen 10% of genes assigned a random number

Hyperparameter-3: Mutation rate 0.1

Genetic Algorithm weight initialisation

Step -4 Eliitness

Top 5 of the population is passed on unhindered to the new generation

Hyperparameter-4: Elitness 5

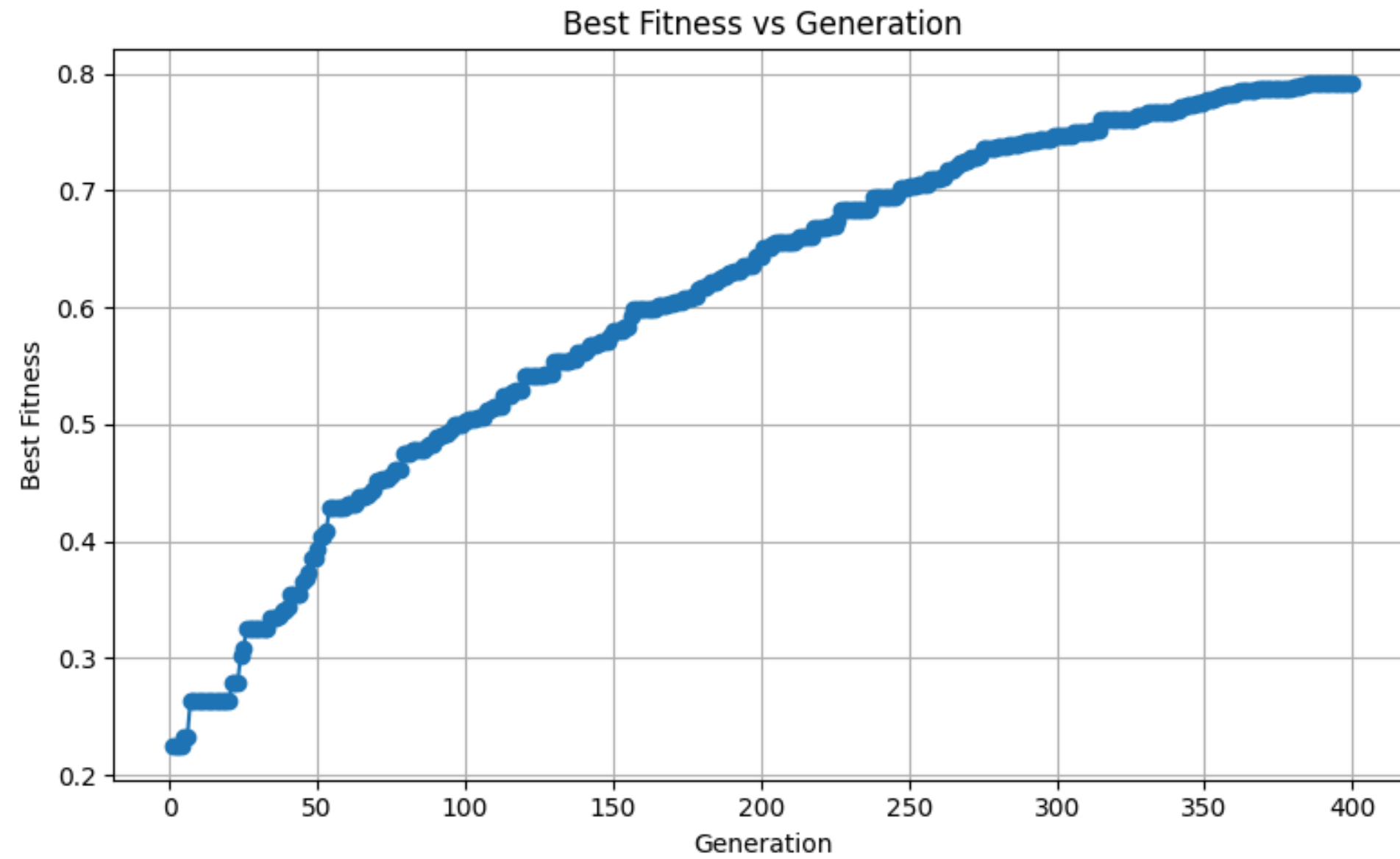
Step -5 Repeat

Process repeated for 400 generations

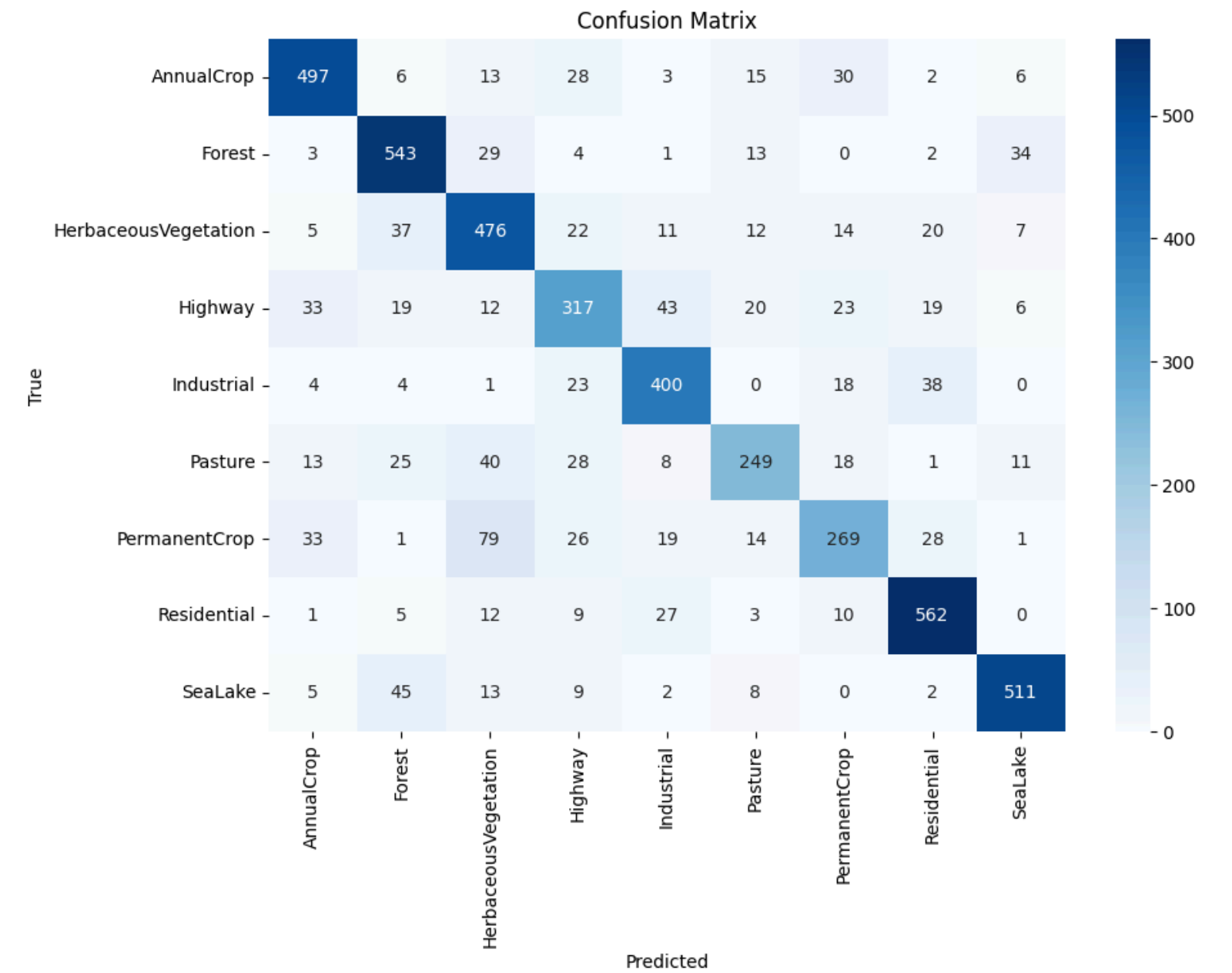
Hyperparameter-5: Generations 400

Results

Fitness vs Generation

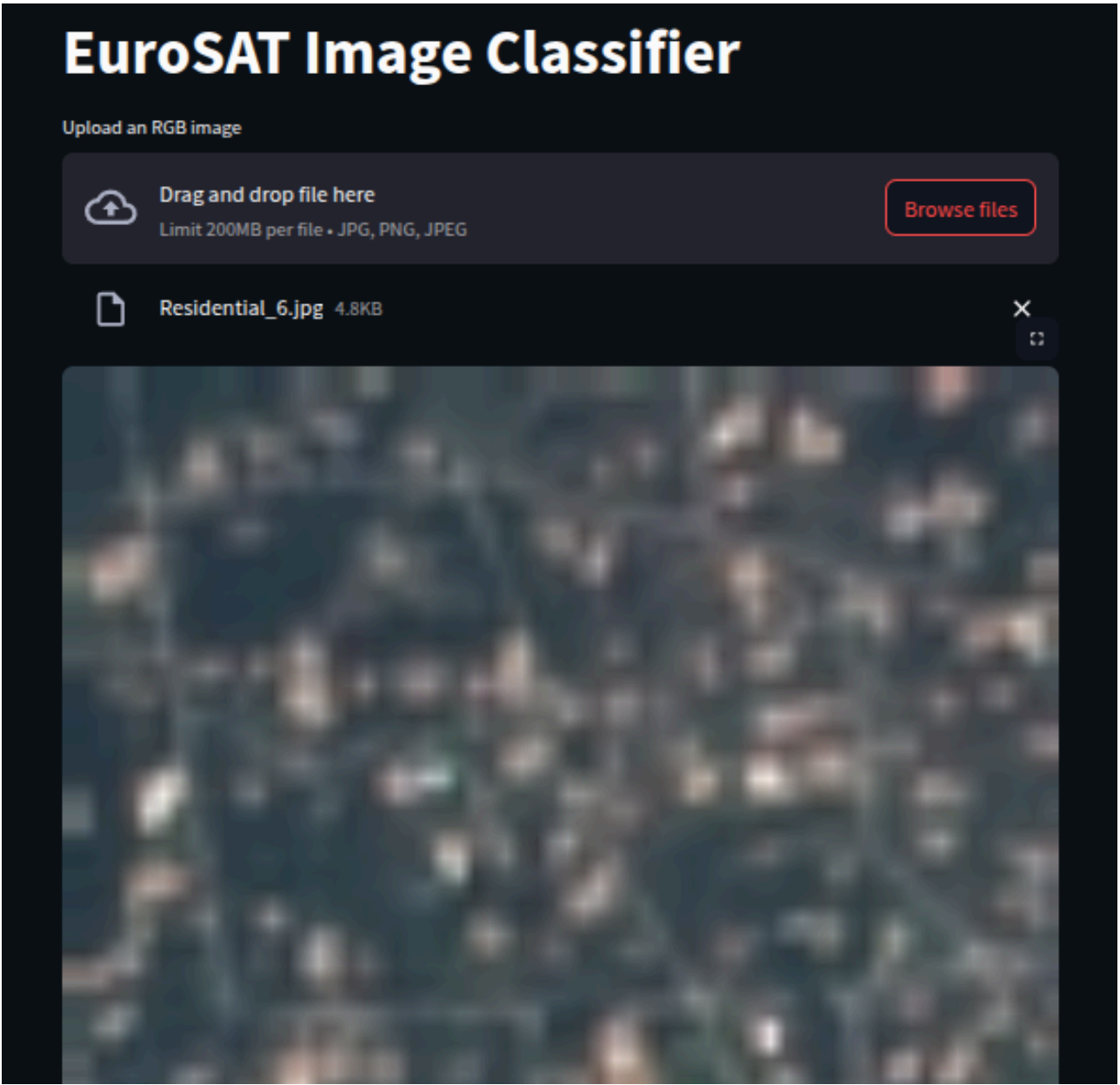


Best Accuracy : 78.04 % (Testing)



Web Interface

Upload Image



Shows top 3 predictions

